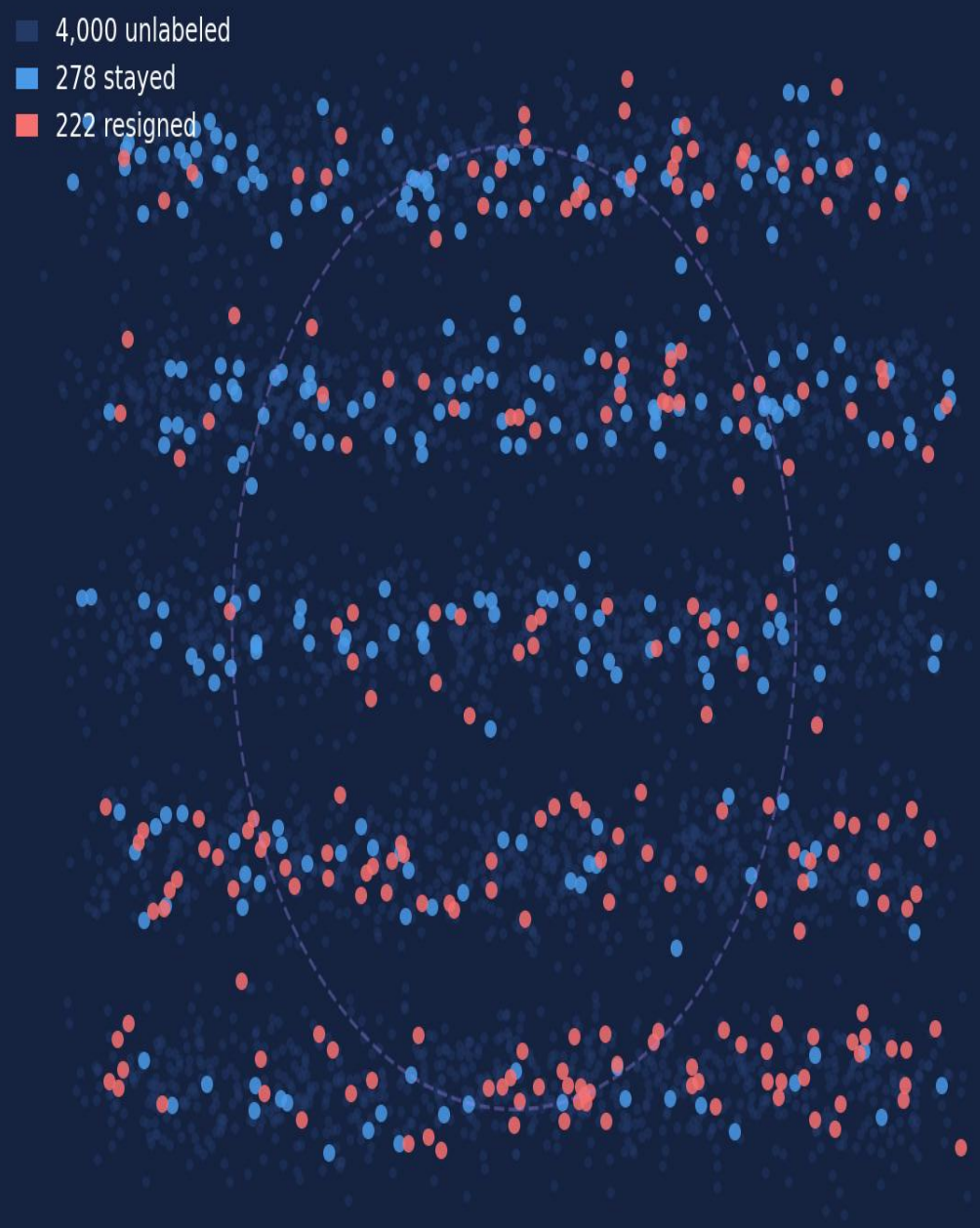
Employee Turnover Prediction

Self-Training · Semi-Supervised

Project 13 · LozanoLsa Industrial ML Portfolio

When 88.9% of your data has no label,
you don't wait — you infer.

Python | scikit-learn | RandomForest | Self-Training



> 4,000

Employee Profiles

500

Labeled Records

0.794

AUC-ROC Score

206

Pseudo-Labels

Luis Lozano | Operational Excellence Manager · Master Black Belt

The Real Problem

Labels only accumulate through departures

A resignation record only exists after someone leaves. Every employee still working is, by definition, unlabeled. The dataset grows with time, but never catches up.

Waiting is a strategy that always arrives too late

Organizations that delay modeling until they have 'enough' labeled data will always be predicting exits that already happened. The risk window has passed.

88.9% of what you know is being discarded

4,000 profiles contain real satisfaction scores, overtime patterns, and workload signals. A purely supervised approach throws all of that away because there is no label.

500 confirmed outcomes · 4,000 active unknowns · Self-Training bridges the gap

Most organizations have far more observations than labels — and they keep collecting the wrong kind of data.

From Labeled Events to Probabilistic Inference

⊘ Old Thinking

— Collect resignation records.

— Train on labeled data only.

— Ignore 4,000 unlabeled profiles.

— Wait for more labels to accumulate.

— Model is always predicting the past.

— Coverage: 11.1% of the workforce.

→

✓ New Thinking

✓ Start with 400 labeled training records.

✓ Score 4,000 unlabeled profiles.

✓ Promote only high-confidence predictions.

✓ Threshold: $P \geq 0.90$ or $P \leq 0.10$.

✓ Expand to 606 training examples.

✓ Evaluate on real labels only. Always.

Self-Training does not invent labels. It promotes only those predictions the base model is already most confident about.

The Data Behind the Problem

HRIS

SAP / Workday

age · years_at_company · job_level

Core employee record — available from day one of hiring.

Payroll

ADP / SAP HR

monthly_salary

Exact gross compensation — updated each payroll cycle.

Survey

Glint / Qualtrics

job_satisfaction · workload

Quarterly pulse surveys — the voice of the employee in the data.

T&A

Time & Attendance

avg_overtime_hours

Monthly aggregation of clock records — proxy for workload-contract mismatch.

What makes this dataset unique

- 7 features — zero preprocessing needed. Random Forest is scale-invariant.
- 500 labeled records (11.1%) — rare enough to matter, enough to bootstrap.
- 4,000 unlabeled rows are not noise. They are signals waiting to be read.
- Satisfied high-salary employees rarely resign — the model learned this cleanly.
- Low satisfaction + high workload = near-100% resignation rate in labeled set.

The label column (resigned) is the rarest variable. Everything else is collected in real-time by operational systems.

Why Self-Training?

"Semi-supervised learning exists because labels are expensive and observations are cheap. Self-Training is the simplest honest answer to that asymmetry."

01 Label efficiency

Supervised training uses 400 examples. Self-Training uses 606. That 51.5% expansion comes from data we already had — salary, satisfaction, overtime — just without a confirmed outcome attached.

02 Confidence-gated expansion

Only predictions at $P \geq 0.90$ or $P \leq 0.10$ enter training. 3,794 employees in the uncertain middle are deliberately excluded. Selectivity protects signal quality.

03 No preprocessing pipeline

Random Forest is scale-invariant and handles mixed numeric types natively. No normalization, no encoding. The model earns its keep by reducing infrastructure complexity.

04 Evaluation stays honest

The 100-record test set was separated before any pseudo-labeling occurred. Metrics reflect performance on real labels only — not the model's own confidence.

The algorithm does not invent knowledge. It extends what the model already believes with high confidence.

Model Results

71.0%	0.794	0.642	70.3%
Accuracy	AUC-ROC	F1 Score	Precision

Top Drivers — Feature Importance

job_satisfaction	0.218
monthly_salary	0.168
workload	0.167
avg_overtime_hours	0.154
years_at_company	0.130
age	0.124
job_level	0.040

Confusion Matrix (n = 100 test records)

	Pred: Stay	Pred: Resign
Actual: Stay	45 ✓	11 ✗
Actual: Resign	18 ✗	**26 ✓**

Recall: 59.1% — Of 44 real resigners, the model flagged 26

Evaluated on 100 real held-out records. Pseudo-labels never entered the test set. AUC-ROC 0.794 — meaningful ranking with label-scarce training data.

From Insight to Action

job_satisfaction 0.218

Satisfaction Signal

→ Quarterly pulse checks for Level 1–2 employees. Flag anyone scoring ≤ 2 for immediate 1:1 with their manager. Do not wait for the annual survey.

monthly_salary 0.168

Compensation Anchor

→ Annual market benchmarking by role and level. Proactive adjustments for employees in the 6–10 year tenure band before they begin job searching.

workload 0.167

Workload Redistribution

→ Trigger HR review when workload self-score ≥ 4 persists for 60+ days. Overloaded employees rarely self-report — the data reports for them.

avg_overtime_hours 0.154

Overtime Cap

→ Structured policy: HR review activated when average overtime exceeds 15 h/month for two consecutive months. Chronic overtime is a contract violation in slow motion.

years_at_company 0.130

Tenure Milestone Review

→ Career path review at Years 3, 5, and 8. These are the departure risk windows where unspoken ambitions collide with perceived ceilings.

Prediction without intervention is just expensive reporting. Each driver maps to a specific retention lever.

What-If Simulator

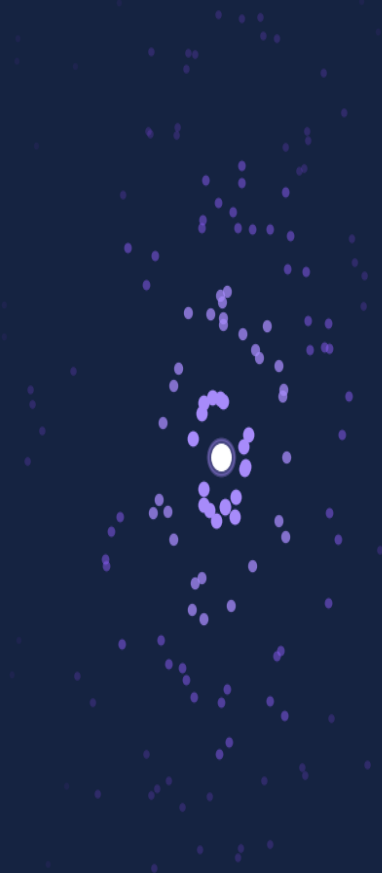
Scenario A · Burnout Operative		Scenario B · Stable Analyst		Scenario C · Manager / High Load	
Age	28	Age	33	Age	42
Tenure	6 yrs	Tenure	3 yrs	Tenure	8 yrs
Level	1 — Operative	Level	2 — Analyst	Level	4 — Manager
Salary	$9,000	Salary	$21,000	Salary	$60,000
Satisfaction	1 / 5	Satisfaction	4 / 5	Satisfaction	3 / 5
Workload	5 / 5	Workload	3 / 5	Workload	5 / 5
Overtime	18 h/mo	Overtime	4 h/mo	Overtime	15 h/mo
P(resign) = 94%		P(resign) = 2%		P(resign) = 34%	

Scenario C insight: A $60,000 salary does not neutralize satisfaction=3 + workload=5 + 15h overtime. Risk sits at 33.5% — a clear HR intervention signal.

Scenario C shows that even a high salary does not offset low satisfaction and sustained overtime for a manager-level employee.

Learning from what you don't yet know.

Self-Training asked the base model one question: what do you already believe, and how confident are you?
Only the most certain answers made it into training. The rest stayed uncertain — and that restraint is what made the model trustworthy.

AUC-ROC 0.794 on a label-scarce HR problem. Not magic — epistemology.

Full dataset (4,500 rows) · app.py simulator · PPTX deck
Challenge the threshold. Set it to 0.80. Does the model improve or collapse?
Because the boundary between labeled and unlabeled has no finish line.